41 questions | **CLO** 4-A | **Midterm** 3/11/2026 · 6:15 pm

ANS Full worked answers: Sections 4-1 and 4-2

الإجابات الكاملة — افتحه بعد حل ملف الأسئلة

Each answer shows the set or the rule, the substitution and the value. The `why` lines name the trap the question was built on.

Part A · Multiple choice (16)

Give the letter and a one-line reason, as in a written exam.

A1 4-1

(b)

Primes 2, 3, 5, 7, 11 occur 1 + 2 + 4 + 6 + 2 = 15 times out of 36: 15/36 = 5/12 ≈ 0.4167. Option (c) uses 11 possible sums as if they were equally likely, which they are not.

A2 4-1

(b)

At least two = exactly two or three: {HHT, HTH, THH, HHH}, 4 of 8 outcomes = 0.5. (a) counts only *exactly* two.

A3 4-1

(c)

0 ≤ P(E) ≤ 1, so a negative value is impossible.

A4 4-1

(b)

The probabilities of all outcomes in S add to 1: k = 1 − 0.70 = 0.30.

A5 4-1

(b)

It is computed from observed frequencies, f/n, so it is empirical.

A6 4-1

(c)

An expert's educated estimate, with no sample space counted and no data table: subjective.

A7 4-1

(b)

$P(\bar{E}) = 1 - P(E) = 0.73$.

A8 4-1

(b)

"At least 2" means 2, 3, 4 or 5 pass, so its complement is 0 or 1 pass: fewer than 2 (that is, at most 1).

A9 4-1

(a)

There are 6 red face cards (J, Q, K of hearts and of diamonds): $6/52 = 3/26$.

A10 4-1

(a)

Every face is greater than 0, so the event is certain. A number greater than 6 is impossible (probability 0).

A11 4-2

(b)

$A \cap B = \{2\}$, so the events are not mutually exclusive: $3/6 + 3/6 - 1/6 = 5/6$. Option (a) forgets to subtract the overlap.

A12 4-2

(b)

Mutually exclusive: $P(A \cap B) = 0$, and $P(A \cup B) = P(A) + P(B) = 0.75$. Option (a) multiplies, as if the events were independent.

A13 4-2

(b)

If they were mutually exclusive, P(A ∪ B) would be 1.1 > 1, which is impossible. So they must overlap, with P(A ∩ B) ≥ 0.1.

A14 4-2

(a)

P(A ∩ B) = P(A) + P(B) − P(A ∪ B) = 0.9 − 0.7 = 0.2.

A15 4-2

(b)

4/52 + 13/52 − 1/52 = 16/52 = 4/13. The queen of clubs is counted in both events, so subtract it once.

A16 4-2

(b)

A only = P(A) − P(A ∩ B) = 0.3.

Part B · Short structured questions (14)

Typical 2–6 mark midterm items. Write S and E in set notation.

B1 4-1

(a) S = {BBBB, BBBG, BBGB, BBGG, BGBB, BGBG, BGGB, BGGG, GBBB, GBBG, GBGB, GBGG, GGBB, GGBG, GGGB, GGGG}, `n(S) = 16`.

(b) E = {BBGG, BGBG, BGGB, GBBG, GBGB, GGBB}, n(E) = 6, so P = 6/16 = `0.375`.

(c) Complement of "no boys" = {GGGG}: 1 − 1/16 = `15/16 ≈ 0.9375`.

(d) 8 outcomes have matching first and last letters: 8/16 = `0.5`.

B2 4-1 4-2

n(S) = 36.

(a) {(2,6), (3,5), (4,4), (5,3), (6,2)}: **5/36 ≈ 0.1389**.

(b) Sums 2, 3, 4 → 1 + 2 + 3 = 6: **6/36 = 1/6 ≈ 0.1667**. "At most 4" includes 4.

(c) Doubles 6, sum 8 five, and (4,4) is in both: (6 + 5 − 1)/36 = **10/36 = 5/18 ≈ 0.2778**.

(d) (1,3), (3,1), (2,4), (4,2), (3,5), (5,3), (4,6), (6,4): **8/36 = 2/9 ≈ 0.2222**.

(e) **0**: impossible, because the largest sum is 12.

B3 4-1

Empirical probability, f/n with n = 100.

(a) 25/100 = **0.25** (b) 0 and 1: (12 + 30)/100 = **0.42** (c) 3 and "4 or more": (18 + 15)/100 = **0.33** (d) only 2 and 3: (25 + 18)/100 = **0.43**

Check: at most 1, exactly 2 and at least 3 cover every student once: 0.42 + 0.25 + 0.33 = 1 ✓.

B4 4-1 4-2

(a) A = {3, 6, 9, 12, 15}; B = {1, 3, 5, 7, 9, 11, 13, 15}; A ∩ B = {3, 9, 15}; A ∪ B = {1, 3, 5, 6, 7, 9, 11, 12, 13, 15}; $\bar{A}$ = {1, 2, 4, 5, 7, 8, 10, 11, 13, 14}; $(A \cup B)^c$ = {2, 4, 8, 10, 14}.

(b) 5/15 + 8/15 − 3/15 = **10/15 = 2/3 ≈ 0.6667** ✓ (n(A ∪ B) = 10).

(c) **No**: A ∩ B = {3, 9, 15} ≠ ∅.

B5 4-1 4-2

(a) 65/200 = **0.325** (b) one cell: 20/200 = **0.10** (c) (65 + 100 − 20)/200 = 145/200 = **0.725** (d) mutually exclusive colleges: (70 + 65)/200 = **0.675** (e) not Engineering and not car = Science-bus + Arts-bus = (40 + 40)/200 = **0.40** (f) **No**: 20 students are both, so P(E ∩ Bus) = 0.10 ≠ 0.

B6 4-2

(a) 0.35 + 0.45 − 0.62 = **0.18** (b) 0.35 − 0.18 = **0.17** (c) 1 − 0.62 = **0.38** (d) A only + B only = 0.17 + 0.27 = **0.44** (or P(A ∪ B) − P(A ∩ B)).

Draw the Venn diagram first: 0.17 | 0.18 | 0.27 | outside 0.38. The four regions add to 1 ✓.

B7 4-2

The brands are mutually exclusive. (a) 0.32 + 0.41 = **0.73** (b) 1 − (0.32 + 0.41 + 0.15) = **0.12** (c) 1 − 0.41 = **0.59**

B8 4-2

(a) Extended addition rule: (50 + 40 + 30 − 15 − 10 − 8 + 5)/100 = **0.92** (b) 1 − 0.92 = **0.08** (c) 50 − 15 − 10 + 5 = 30, so **0.3**. Add 5 back because the triple overlap was subtracted twice.

B9 4-1

Those three outcomes are not equally likely, so classical probability does not apply to them. The equally likely sample space is S = {HH, HT, TH, TT}, and P(exactly one head) = 2/4 = **0.5**.

B10 4-2

The king of hearts is counted twice; the events are not mutually exclusive. Correct: 4/52 + 13/52 − 1/52 = **16/52 = 4/13**.

B11 4-1

P(E) = 3P($\bar{E}$) = 3(1 − P(E)), so 4P(E) = 3 and P(E) = **0.75**.

B12 4-2

English or French = 120 − 25 = 95, so both = 70 + 45 − 95 = 20.

(a) **20/120 = 1/6 ≈ 0.1667** (b) (95 − 20)/120 = **75/120 = 5/8 ≈ 0.625** (c) (45 − 20)/120 = **25/120 = 5/24 ≈ 0.2083**

B13 4-1

The product is odd only when both numbers are odd: 3 × 3 = 9 outcomes. P(even) = 1 − 9/36 = **27/36 = 3/4 ≈ 0.75**.

B14 4-2

(a) **Not mutually exclusive**: a Science student can be taking STAT 2101. (b) **Mutually exclusive**: odd = {1, 3, 5} and greater than 5 = {6}, so the intersection is empty. (c) **Mutually exclusive**: spades are black.

Part C · Long questions (5)

Midterm-style, multi-part. Marks in brackets.

C1 4-1 4-2 **[12 marks]**

1. **Empirical**: it comes from observed frequencies (f/n).

2. 34/200 = **0.17**

3. Not mutually exclusive (A− = 10): (66 + 34 − 10)/200 = **0.45**

4. Not O and not Rh+, i.e. A−, B− or AB−: (10 + 6 + 4)/200 = **0.10**

5. Addition rule 1 (a donor has one type): (66 + 30)/200 = **0.48**

6. 1 − 20/200 = **0.90**

7. **No**: 14 donors are O−.

C2 (4-1) (4-2) **[8 marks]**

1. S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}, **n(S) = 12**.

2. {H2, H4, H6}: 3/12 = **0.25**

3. 6/12 + 6/12 − 3/12 = **9/12 = 0.75**

4. {T5, T6}: **2/12 ≈ 0.1667**

5. **No**: H6 belongs to both.

C3 (4-1) (4-2) **[10 marks]**

1. **36** ordered pairs (red, green); the dice are fair, so each pair has probability 1/36.

2. **18/36 = 1/2 ≈ 0.5**

3. sum > 9 has 6 outcomes and odd-and->9 (sum 11) has 2: (18 + 6 − 2)/36 = **22/36 = 11/18 ≈ 0.6111**

4. red 6: 6 outcomes; sum 7: 6 outcomes; (6, 1) is in both: (6 + 6 − 1)/36 = **11/36 ≈ 0.3056**

5. Mutually exclusive sums: (6 + 2)/36 = **8/36 = 2/9 ≈ 0.2222**; P(lose) = **28/36 = 7/9 ≈ 0.7778**

C4 (4-1) (4-2) **[8 marks]**

1. 12/50 = **0.24** **2.** (10 + 5)/50 = **0.30** **3.** (9 + 14 + 5)/50 = **0.56** (mutually exclusive classes) **4.** 1 − 14/50 = **0.72**

5. **No**: 30 falls inside the 26–33 class, and the table doesn't show how those 14 ages split around 30.

C5 (4-2) **[10 marks]**

1. A only 0.28 · both 0.12 · B only 0.23 · outside 0.37 (sum = 1 ✓).

2. **0.63** **3.** **0.37** **4.** everything except A only: 1 − 0.28 = **0.72** **5.** 0.28 + 0.23 = **0.51**

6. **No**: A ∩ B is part of B, so P(A ∩ B) ≤ P(B) = 0.35.

Part D · 20-minute mini-quiz (16 marks)

Time yourself, closed book, before opening the answers.

D1 4-1

The **set of all possible outcomes** of a probability experiment (one die: {1, ..., 6}). An event with **two or more outcomes** (an even number: {2, 4, 6}).

D2 4-1

Exactly one: {HHT, HTH, THH}, **3/8**. At most one: no tails or one tail, **4/8 = 0.5**. No tails: {HHH}, **1/8**.

D3 4-2

(a) **0.65** (b) 0.25 + 0.40 − 0.10 = **0.55**

D4 4-1 4-2

(18 + 12 − 7)/30 = **23/30 ≈ 0.7667**. Male without glasses = the complement of (female or glasses) = **7/30 ≈ 0.2333**. Check: 12 males, 5 with glasses, so 7 without ✓.

D5 4-1

Sums 10, 11, 12: 3 + 2 + 1 = 6, **6/36 = 1/6 ≈ 0.1667**.

D6 4-2

They share **no outcomes** (A ∩ B = ∅), e.g. "even" and "odd" on one die. Rule: **P(A ∪ B) = P(A) + P(B)**.

Built to the STAT 2101 Fall 2026 course outline (CLO4-A: calculate probabilities using basic probability principles). Sections 4-1 Sample Spaces and Probability and 4-2 The Addition Rules only. Every numeric answer was computed by code from the question data.